Extended reading — Week 11: Contextual Bandits and Off-Policy Evaluation

November 8, 2026

Read with a question in mind

Start with the week's teaching guide and its small worked example. Then use these readings to connect the algorithm to its assumptions and evidence. The explanations and investigations below are course commentary, not quotations. Read the source before reusing a theorem or a reported result. For each paper, record the objective, observation model, claimed result, conditions and one condition your local experiment does not reproduce.

1. Auer, Cesa-Bianchi and Fischer (2002): Finite-time Bandit Analysis

Primary source or book record: https://homes.di.unimi.it/~cesabian/Pubblicazioni/ml-02.pdf.

What to learn. UCB1 gives a finite-time expected-regret guarantee for bounded stationary rewards. The leading per-arm contribution is proportional to $\log T/\Delta_a$.

What the result requires. The suboptimal-arm count is bounded above by a threshold plus bad-event contributions. Do not reverse the inequality and treat the threshold as a required minimum for each suboptimal selection.

Notebook investigation. Use `exercise_1_ucb_bonus`. Annotate the proof into initial pulls, underexplored pulls and concentration failures; identify the assumption behind each term.

2. Li et al. (2010): Contextual-bandit News Recommendation

Primary source or book record: https://arxiv.org/abs/1003.0146.

What to learn. Disjoint and hybrid linear models show different ways to share statistical information across available actions. Randomized logging supports a replay evaluation.

What the result requires. An implementation with a tuning coefficient is not automatically the determinant-based confidence set from later theory. With K independent d-dimensional arm models the joint parameter dimension is Kd, not d.

Notebook investigation. Use `exercise_4_linucb_regret`. Inspect whether a single update changes one arm model or a shared model. Do not claim a linear-reward theorem for a sigmoid reward mean without additional analysis.

3. Agrawal and Goyal (2013): Thompson Sampling for Contextual Bandits with Linear Payoffs

Primary source or book record: https://arxiv.org/abs/1209.3352.

What to learn. Studies a linear contextual Thompson-sampling algorithm and its frequentist regret under explicit feature and noise assumptions.

What the result requires. A common shared linear parameter differs from independent arm-specific parameters. A sampled parameter vector chooses an action but does not by itself give the marginal probability of choosing that action.

Notebook investigation. For three candidate features, sample one parameter vector and score all three. Repeating the full sampling-and-selection process approximates action frequencies; one Gaussian density value does not.

4. Dudik, Langford and Li (2011): Doubly Robust Policy Evaluation and Learning

Primary source or book record: https://arxiv.org/abs/1103.4601.

What to learn. Combines a model-based target-policy value with an importance-weighted residual correction. Correct propensity or reward-model conditions give the double-robustness property.

What the result requires. Overlap and appropriate fitting/evaluation separation still matter. A poor model can inflate variance, so there is no unconditional dominance over IPS. Consistency is not the same as finite-sample unbiasedness. With adaptive logging, the relevant recorded probability is conditional on the decision history; an IID-row variance formula needs a separate justification. The supplied empirical figures instead use independent repetitions of complete logs and separate nuisance fits, conditional on one fixed reward model and target.

Notebook investigation. Use `exercise_6_dr_softmax`. Set the fitted reward to zero and check recovery of IPS. Then perturb reward predictions while keeping logged propensities exact.

5. Joachims, Swaminathan and de Rijke (2018): BanditNet

Primary source or book record: https://www.cs.cornell.edu/~tj/publications/joachims_etal_18a.pdf.

What to learn. Addresses learning deep policies from logged feedback with a counterfactual objective and a tractable optimisation formulation.

What the result requires. Propensity overfitting means exploiting weaknesses of the empirical counterfactual objective, not merely estimating a logger from data. This is one method among many, not the only published way to learn offline.

Notebook investigation. Split logs before tuning a policy. Compare its training objective with held-out IPS, SNIPS and DR, keeping reward transformations and clipping thresholds explicit.

6. Saito and Joachims (2022): Off-Policy Evaluation for Large Action Spaces via Embeddings

Primary source or book record: https://arxiv.org/abs/2202.06317.

What to learn. Marginalises importance weights through action embeddings to exploit structure and potentially reduce variance.

What the result requires. Unbiasedness needs conditions connecting embeddings, actions and rewards, including the absence of a direct action effect in the relevant formulation. There is no universal replacement of K by embedding dimension in a variance formula. A residual check cannot prove sufficiency.

Notebook investigation. Construct two actions with the same embedding but different expected rewards. Explain which assumption fails and why a lower-variance estimate can still be biased.

7. Bottou et al. (2013): Counterfactual Reasoning and Learning Systems

Primary source or book record: https://jmlr.org/papers/v14/bottou13a.html.

What to learn. Studies counterfactual reasoning in learning systems and the relationship between intervention design, data collection and evaluation.

What the result requires. A versioned event record must preserve the actual probability of the executed decision. Different logger versions may coexist in a valid analysis; multiplying every row by one assumed probability is the failure.

Notebook investigation. Draw a decision-to-outcome timeline. Label decision probability, model version, eligibility and outcome delay. Show how excluding immature observations can change the evaluated population.

8. Bietti, Agarwal and Langford (2021): A Contextual Bandit Bake-off

Primary source or book record: https://jmlr.org/papers/v22/18-863.html.

What to learn. Compares practical oracle-based algorithms using supervised datasets converted into bandit tasks. In this study an optimism-based method performs best overall, with a greedy baseline surprisingly competitive and an Online Cover variant also strong.

What the result requires. This is not evidence that a cover-softmax plus DR recipe universally wins on production logs. Dataset conversion, model class, tuning and the definition of aggregate performance matter.

Notebook investigation. Read one result table with its experimental protocol. Identify whether the comparison measures online regret, supervised error or offline estimator quality before applying it to the workbook.

Synthesis: evidence you should be able to show

1. Reproduce one numerical calculation before running a long experiment.
2. Name the precise objective, sampling distribution and terminal convention.
3. Compare the local implementation with one source: list a shared mechanism and one simplification that prevents calling it a full reproduction.
4. Separate an algebraic identity, an assumption-dependent theorem and an empirical result. A plot alone does not prove a theorem.
5. Save the notebook with seed, parameters, axes, denominators and a short explanation of an unexpected result. Keep evaluation data separate from tuning.

Return to the teaching guide's notebook investigation for the local function names, expected checks and interpretation questions. Source code and notebooks are available together in the course repository; the web reader is the entry point and the downloaded notebook is the execution workspace.

Statistics reading bridge

A defined estimate can still have almost no observed overlap. Read estimator bias and bootstrap resampling together. Inspect zero-total-weight resamples explicitly, and separate a conditional fixed-model interval from uncertainty across the complete nuisance-fitting and policy-selection pipeline.

Checked textbook locations: Siegrist § § 4.1, 7.1, PDF pp.439, 896; Ross § 15.3, PDF p.683. https://stats.libretexts.org/@go/page/10189. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.

Read the empirical comparison carefully. The reference figures keep the latent environment and target fixed, regenerate whole adaptive logs and independent uniform fitting samples, and use one shared 50,000-context Monte Carlo value with its MC SE. A numerical reference is not exact ground truth. For SWITCH, the plotted identity is empirical MSE equals squared empirical mean error plus variance with denominator R, all in squared reward units. It is not an assertion that the population bias or variance was known. Reconstruct the identity for errors $(-0.1, 0.1, 0.2)$: $1/50 = 1/225 + 7/450$. Then explain how changing the latent model between repetitions would mix across-environment variability with fixed-target estimation error. A threshold that reduces error on one finite fixture need not be best on another.